Evaluate the Current System Landscape

Given a set of business requirements, identify the current system landscape and determine what standards, limitations, boundaries and protocols exist.

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After studying this topic, you should be able to:

- Identify the standards, protocols, boundaries, and limitations for various integration patterns.
- Identify the standards, protocols, boundaries, and limitations for the given business scenarios.

Introduction

A system landscape may consist of remote systems that must comply with certain **standards and protocols** required for integration. In addition, certain **boundaries or limitations** could also affect the integration of systems. For instance, **synchronous Apex SOAP callouts** require the remote system to be compatible with the **web service standards** supported by Salesforce, which are **WSDL 1.1, SOAP 1.1, WSI-Basic Profile 1.1, and HTTP**. And the **maximum number of callouts** in an Apex transaction cannot exceed **100**.

Standards, Protocols, Boundaries, and Limitations

Standards

A remote system should be compatible with the standards required for integration. For example, for **outbound messaging**, a **remote system** should be able to implement a **listener**.

Protocols

Certain protocols are applicable for implementing integrations. For example, **Streaming API** requires the user interface to use a **JavaScript-based implementation of the Bayeux protocol (currently CometD)**.

Boundaries

Certain boundaries may be applicable to an integration. For example, when using the **SOAP API**, a transaction **cannot span** across multiple API calls.

Limitations

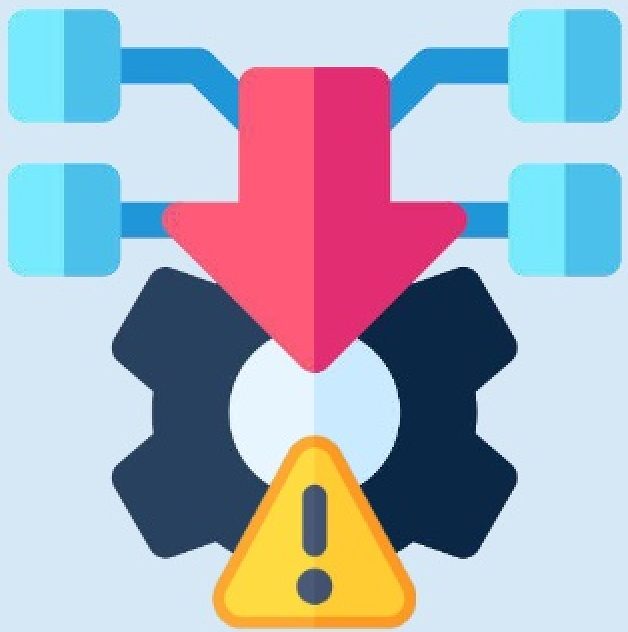
Certain **governor limits** may be applicable to integration solutions. For example, the number of **Apex callouts** in a transaction **cannot exceed 100**.

Standards, Limitations, Boundaries & Protocols

An integration solution may require a remote system to be compatible with certain **standards and protocols** and may be affected by certain **boundaries or limitations**.

REMOTE PROCESS INVOCATION - REQUEST & REPLY

For **synchronous Apex SOAP callouts**, the remote system must be compatible with the **web service standards** supported by Salesforce, which are **WSDL 1.1, SOAP 1.1, WSI-Basic Profile 1.1, and HTTP**. For **Apex HTTP callouts**, the endpoint must be able to receive **HTTP calls** and be **accessible over the public Internet**. If a **middleware** participates in the integration, it must support **synchronous transport protocols**. For **External Services**, the externally hosted service must be a **RESTful service**. Various governor limits exist for **synchronous Apex callouts**. For example, the **total number of callouts** in a transaction cannot exceed **100** and the **maximum cumulative timeout** for all callouts in a transaction is **120 seconds**.



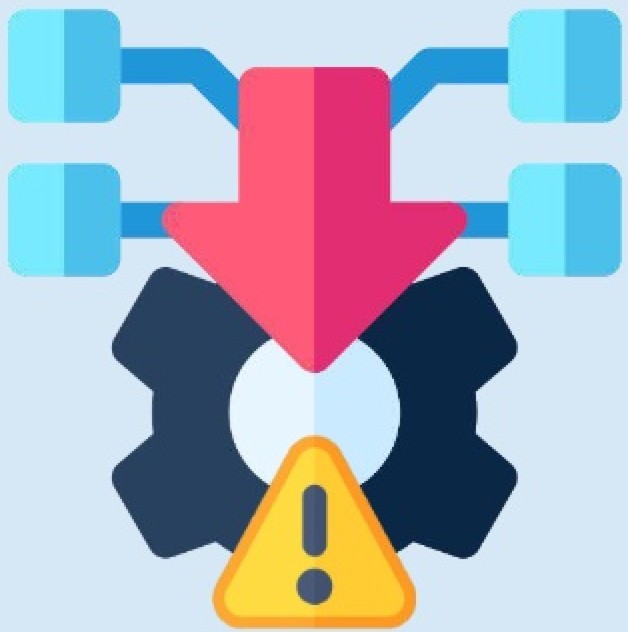
Standards, Limitations, Boundaries & Protocols

An integration solution may require a remote system to be compatible with certain **standards and protocols** and may be affected by certain **boundaries or limitations**.

REMOTE PROCESS INVOCATION - FIRE & FORGET

For asynchronous **Apex REST and SOAP callouts**, the remote system must be compatible with the same standards that apply to synchronous callouts. For **Outbound Messaging**, the remote system must be able to implement a **listener** that can receive **SOAP messages** in the predefined format sent by Salesforce. The remote listener must participate in a **contract-first implementation**, where the contract is supplied by Salesforce.

Various governor limits exist for **asynchronous Apex callouts**. For example, the **maximum number of Apex jobs** that can be added to the queue with **System.enqueueJob** is **1**.



Standards, Limitations, Boundaries & Protocols

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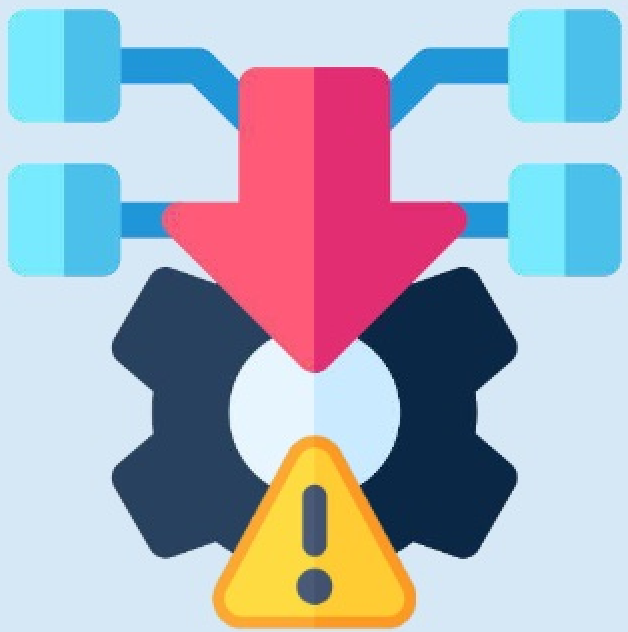
BATCH DATA SYNCHRONIZATION

For **Change Data Capture**, the remote system must have an **integration app** for **receiving events and performing updates** in the external system. Using a **third-party ETL tool** for **Salesforce data replication** requires the use of **Salesforce SOAP API or Bulk API**.

The **HTTPS protocol** should be used for making calls to the Salesforce API. If a **middleware** is used with the ETL tool, it should support **asynchronous transport protocols**.



Standards, Limitations, Boundaries & Protocols

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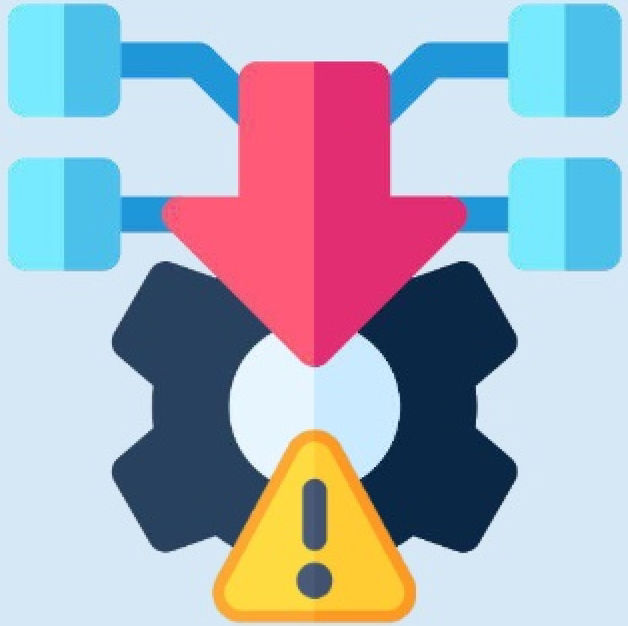
REMOTE CALL-IN

For using **REST/SOAP API** or **Apex REST/web service**, the remote system must be capable of implementing a client capable of invoking the **API or Apex REST/web service**.

When using **SOAP API**, a transaction cannot span across multiple API calls. But it's possible for a single API call to affect multiple objects. When using **REST API**, **REST API composite resources** can be used to make a series of updates in one API call. The number of records that can be processed per API call is limited to **200**. Bulk API 2.0 can be used for more than 2,000 records.

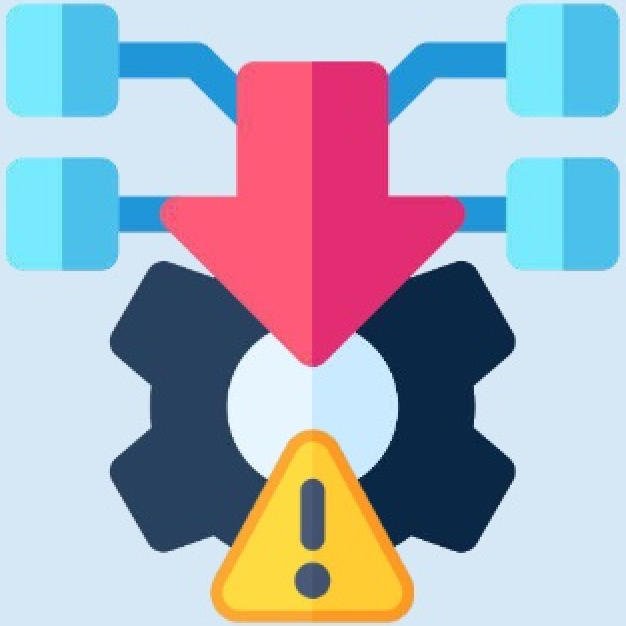


Standards, Limitations, Boundaries & Protocols

An integration solution may require a remote system to be compatible with certain **standards and protocols** and may be affected by certain **boundaries or limitations**.

UI UPDATE BASED ON DATA CHANGES

For using the **Streaming API** to stream notifications, the user interface must use a **JavaScript-based implementation** of the **Bayeux protocol (currently CometD)**. The **HTTPS protocol** is recommended for connection. However, the **delivery and order of notifications aren't guaranteed**, and they aren't generated from record changes made by **Bulk API**.

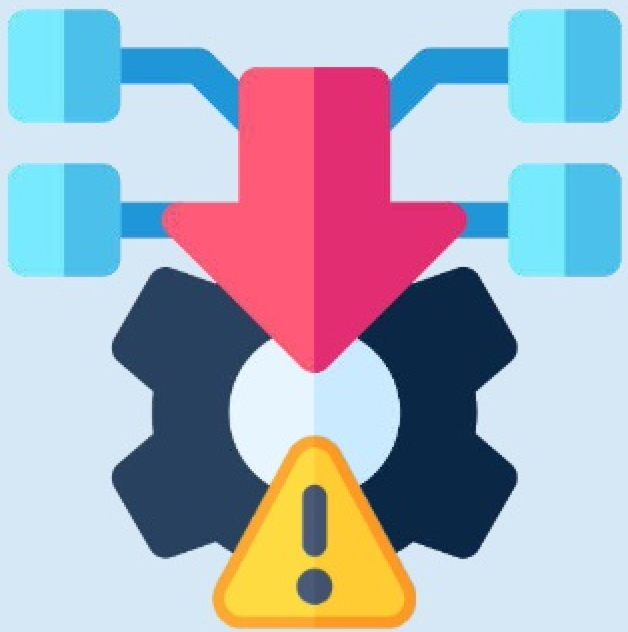


Standards, Limitations, Boundaries & Protocols

An integration solution may require a remote system to be compatible with certain **standards and protocols** and may be affected by certain **boundaries or limitations**.

DATA VIRTUALIZATION

For setting up **Salesforce Connect** for a remote system, the external system must support one of the available adapters, such as **OData 2.0**, **OData 4.0/4.01**, or **cross-org adapter**. For Request & Reply solutions, the remote system must be able to receive **web service or HTTP calls**, depending on the solution. There are **no callout limits** imposed on OData adapters.



Business Scenario

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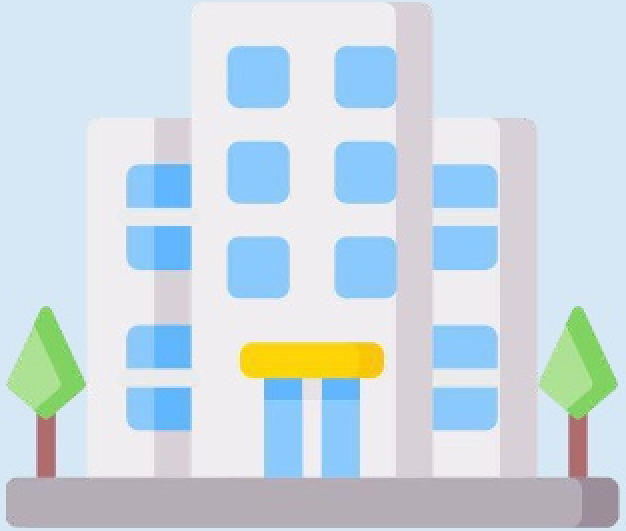
Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Harvest is a company that produces and sells organic food products in the United States. It uses Salesforce and multiple remote systems, such as an ERP system, an SCM (Supply Chain Management) system, and a TPS (Transaction Processing System).

To streamline the existing business processes, the company would like to integrate the systems with one another. The solution architect needs to analyze the system landscape for various requirements and identify the standards, limitations, boundaries and protocols that could affect the solutions.

Requirements & Solutions

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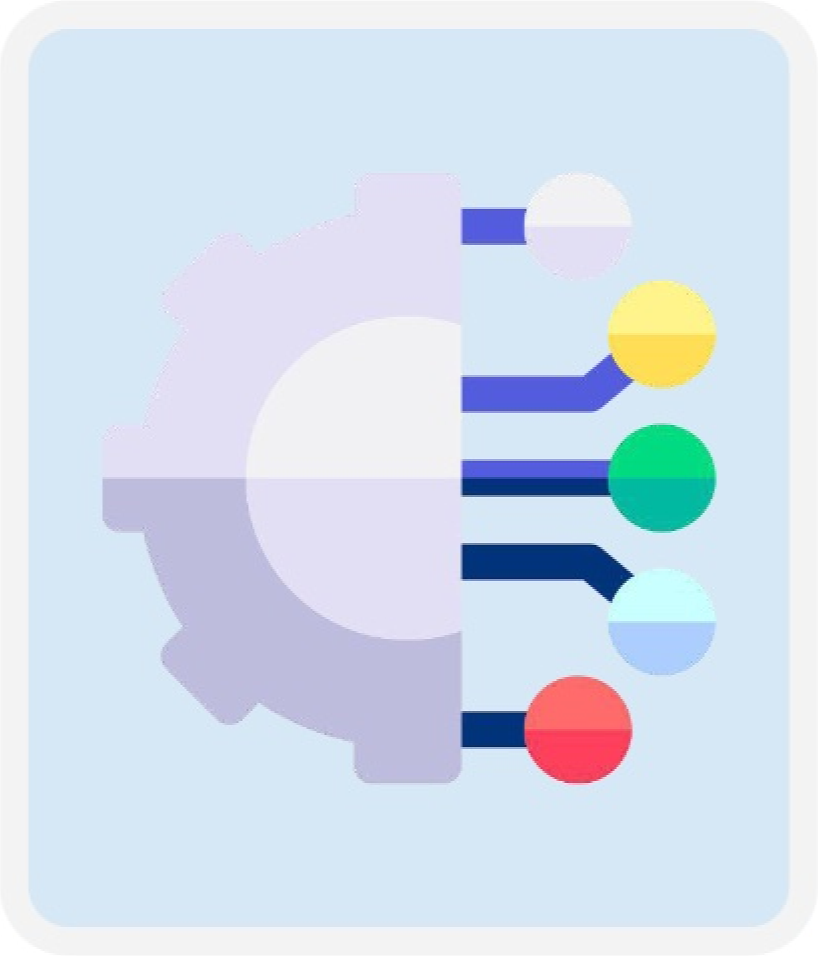
Requirement 1

SCENARIO

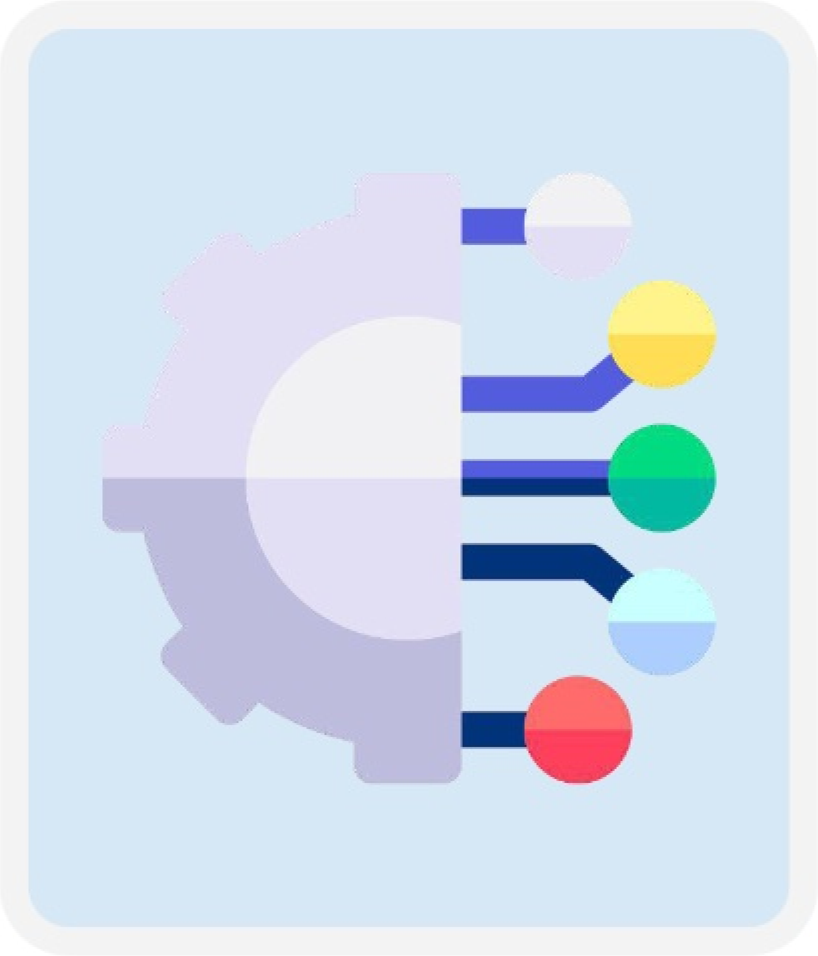
When an order is placed by a customer on the company's website, it is created and processed automatically in the Transaction Processing System (TPS). The TPS must then send the order data to Salesforce. When Salesforce receives the data from the TPS, an Order record should be created based on information that needs to be extracted from the order data. The solution architect must identify any relevant considerations for implementing a Remote Call-In solution for this requirement.



Solution 1

SOLUTION

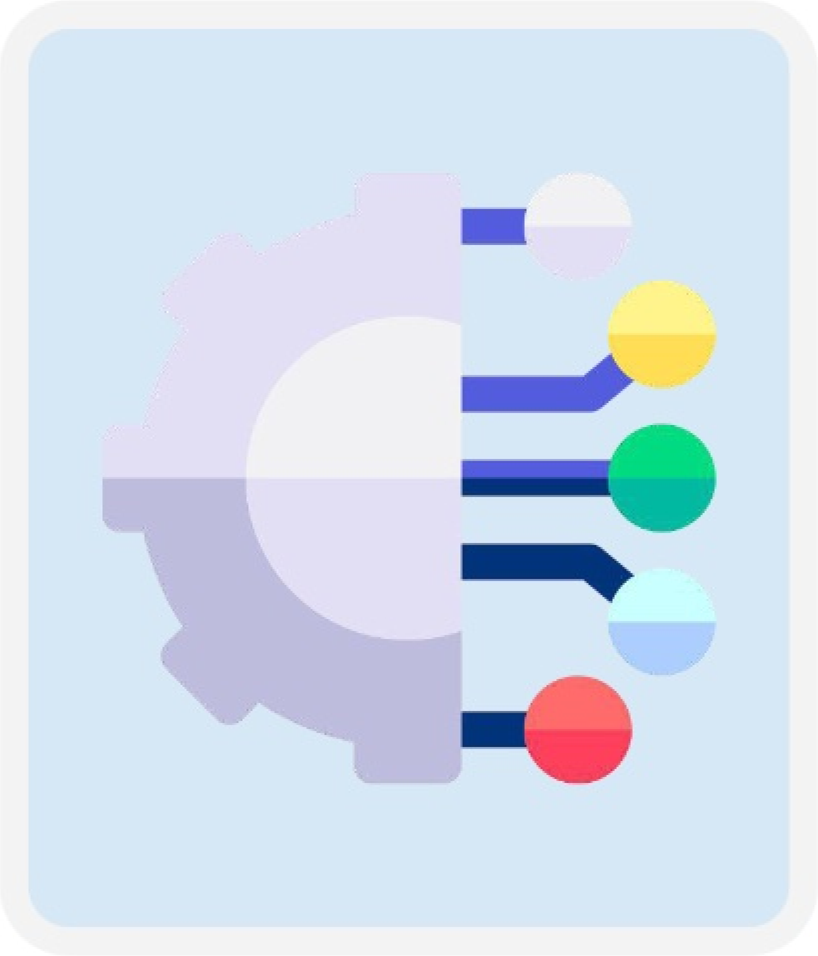
An **Apex REST or web service** can be utilized for this use case. It can receive the order data from the TPS, parse data to extract the necessary information, and create an order record in Salesforce automatically. However, the TPS must **support invoking the Apex REST or web service**, and **additional Apex code** would need to be maintained in Salesforce.



Requirement 2

SCENARIO

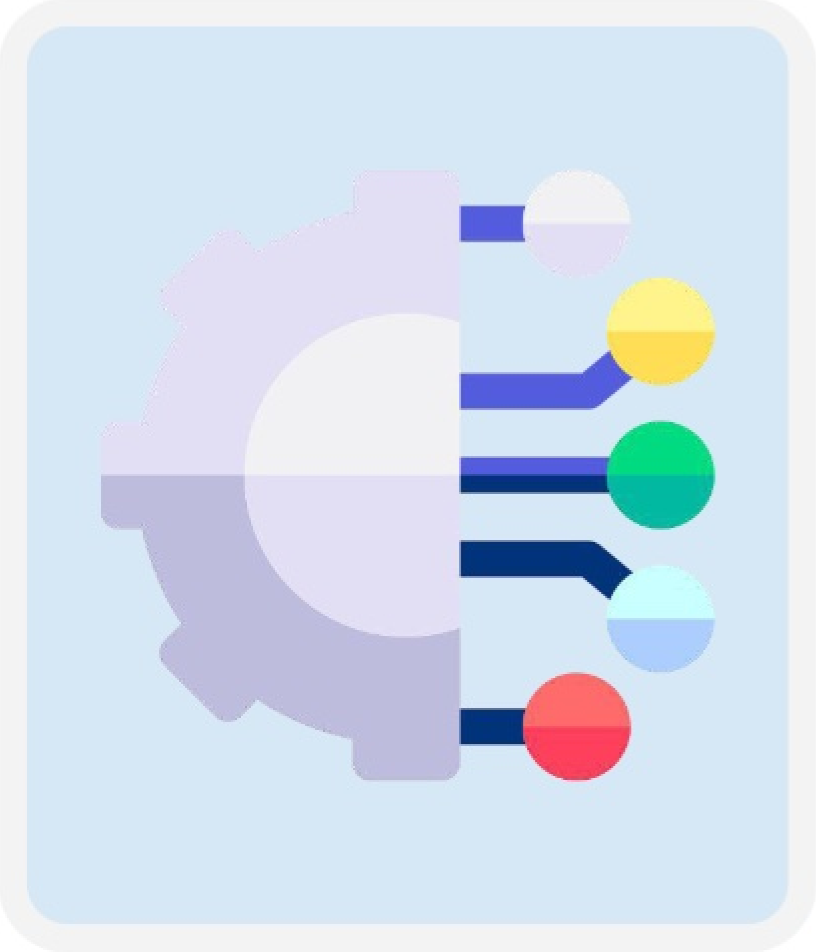
The SCM (Supply Chain Management) system stores information about storage and movement of raw materials required by the company to manufacture food products. Certain Salesforce users should be able to access this information in real-time using a Lightning web component created by a Salesforce developer. The architect must identify any standards, limitations, boundaries and protocols that must be considered for the use of HTTP callouts for this use case.



Solution 2

SOLUTION

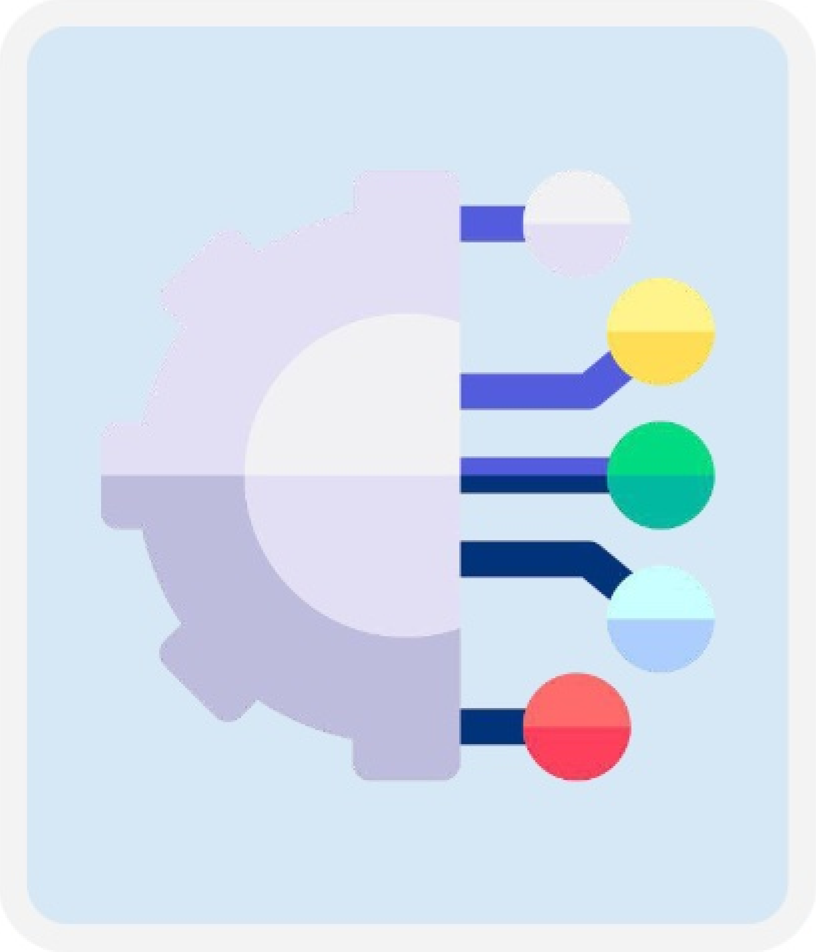
For **synchronous Apex HTTP callouts**, the SCM system must be able to receive **HTTP calls** and be **accessible by Apex over the public Internet**. If a **middleware** participates in the integration, it must support **synchronous transport protocols**. Various governor limits that exist for **synchronous Apex callouts must be considered**. For example, the **total number of callouts** in a transaction initiated by the Lightning web component should not exceed **100**.



Requirement 3

SCENARIO

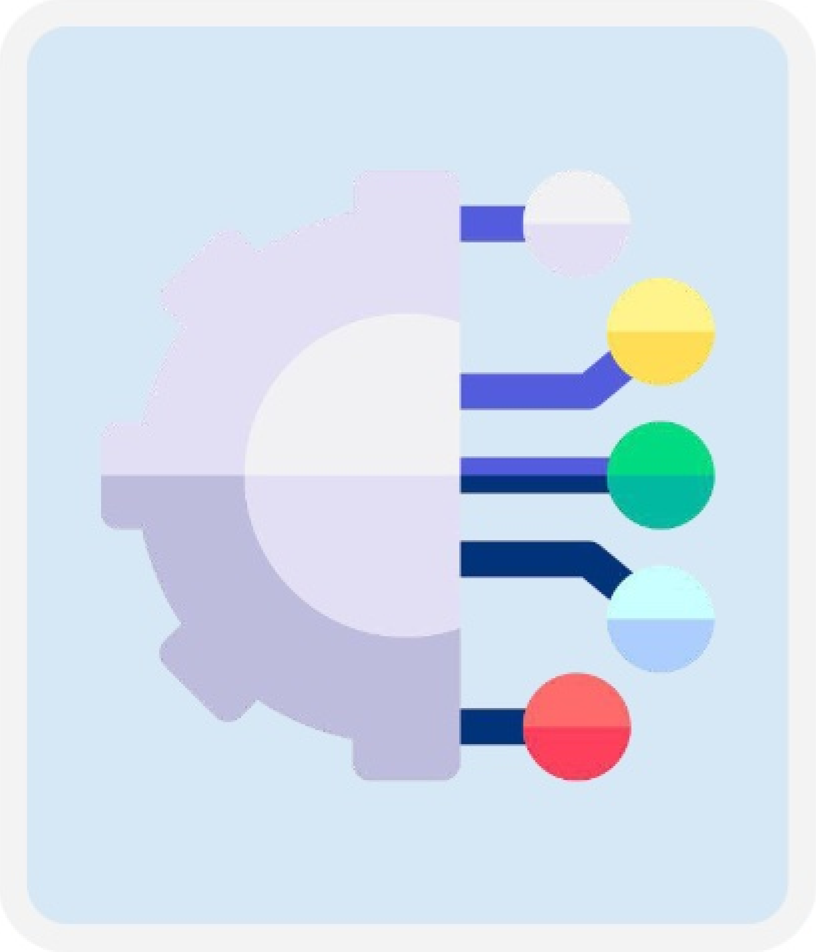
When a new invoice is created in the TPS, any Salesforce users who have access to a Lightning web component should be able to see a message that shows invoice information. The use of Streaming API is being considered for this requirement. The architect must outline any related protocols and limitations.



Solution 3

SOLUTION

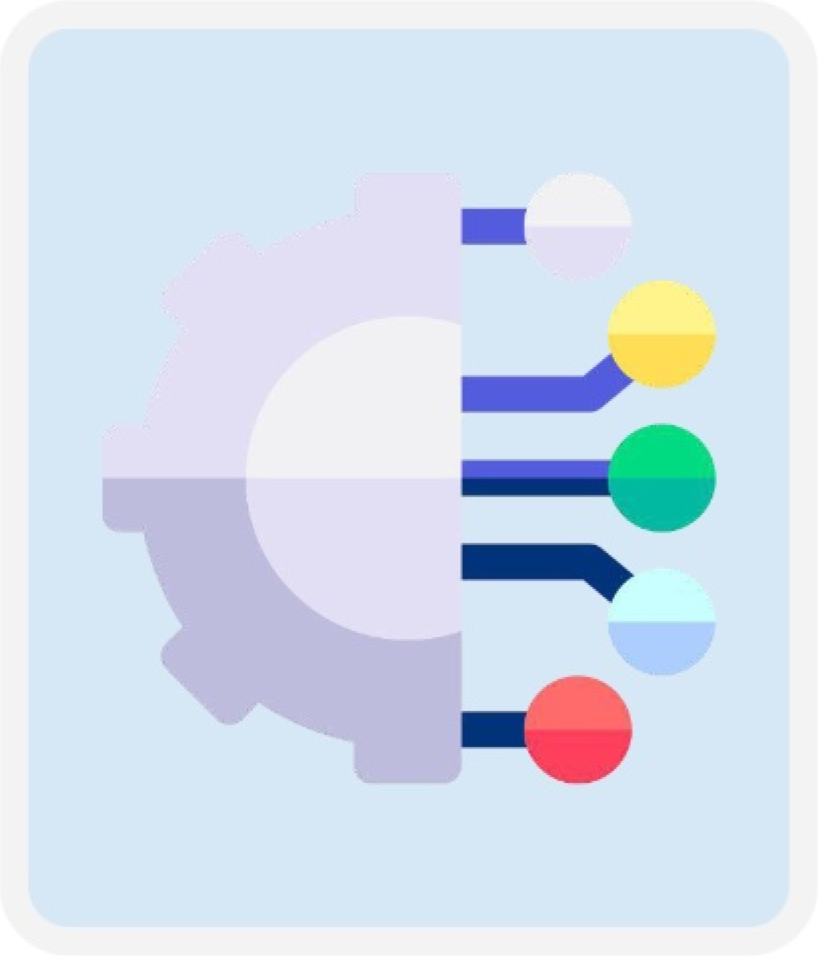
For using the **Streaming API** to stream notifications, the Lightning web component must use a **JavaScript-based implementation** of the **Bayeux protocol (currently CometD)**. The **empAPI** component can be used since it provides access to methods for subscribing to a streaming channel and listening to event messages. However, it is important to note that the **delivery and order of notifications aren't guaranteed.**



Requirement 4

SCENARIO

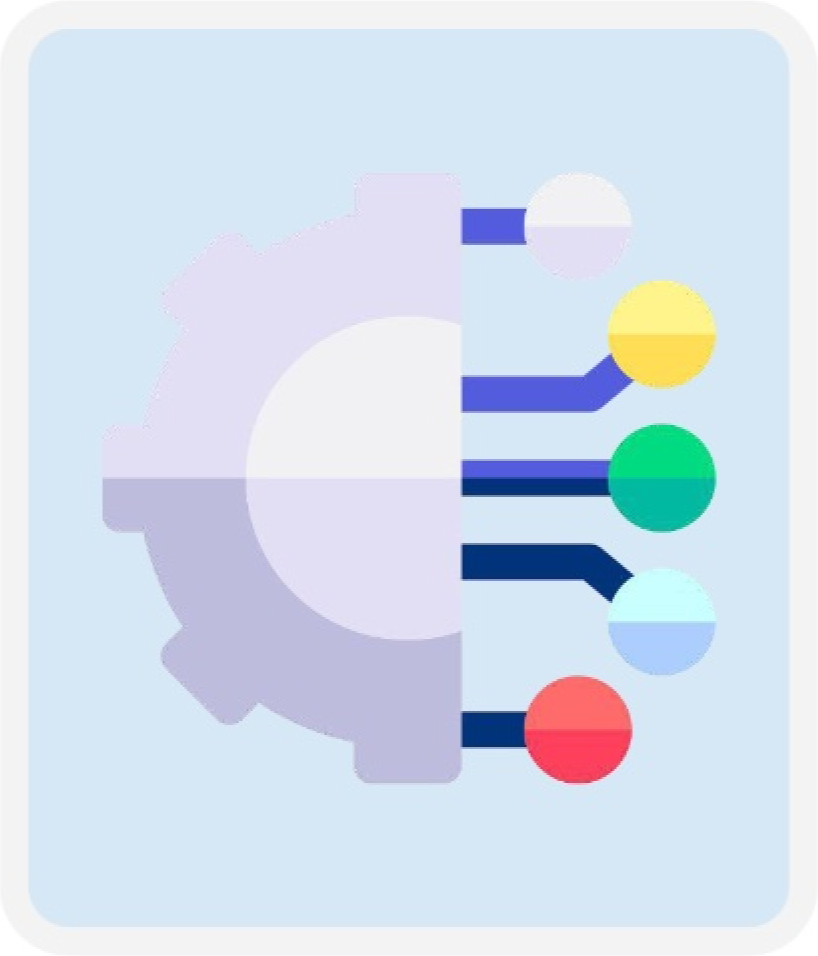
The SCM system stores data about warehouses used by the company. This data needs to be exposed in Salesforce using Salesforce Connect. Various Salesforce users in the sales department will require access to this data through list views, record detail pages, and reports. A flow will also need to access the external object that represents warehouses. The solution architect needs to identify any standards and limitations that should be considered for setting up Salesforce Connect for this requirement.



Solution 4

SOLUTION

For setting up **Salesforce Connect**, the SCM system must support the **OData 2.0/4.0/4.01 adapter**. Salesforce Connect access external data real-time via web service callouts. When using OData adapters, **no callout limits** are imposed. However, it is important to consider any limits that the external data source itself may have.



Learn More

- [Integration Patterns and Practices]
- [Trade-off - Wiki]